Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Let $\mathrm{A}(4,7,8), \mathrm{B}(2,3,4), \mathrm{C}(2,5,7)$ be the vertices of a triangle ABC. The length of internal bisector of $\angle \mathrm{A}$ is

(A) $\frac{\sqrt{34}}{2}$

(B) $\frac{3}{2}\sqrt{34}$

(C) $\frac{2}{3}\sqrt{34}$

(D) $\frac{\sqrt{34}}{3}$

Q.2

The locus of a point, such that the sum of the squares of its distances from the planes $x+y+z=0, x-z=0$ and $x-2y+z=0$ is 9, is

(A) $x^2+y^2+z^2=3$

(B) $x^2+y^2+z^2=6$

(C) $x^2+y^2+z^2=9$

(D) $x^2+y^2+z^2=12$

Q.3

A variable plane passes through a fixed point $(1,2,3)$. The locus of the foot of the perpendicular from the origin to this plane is given by

(A) $x^2+y^2+z^2-14=0$

(B) $x^2+y^2+z^2+x+2y+3z=0$

(C) $x^2+y^2+z^2-x-2y-3z=0$

(D) None of these

Q.4

If Q is the image of the point $\mathrm{P}(2,3,4)$ under the reflection in the plane $x-2y+5z=6$, then the equation of the line PQ is

(A) $\frac{x-2}{-1}=\frac{y-3}{2}=\frac{z-4}{5}$

(B) $\frac{x-2}{1}=\frac{y-3}{-2}=\frac{z-4}{5}$

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(C) $\frac{x-2}{-1}=\frac{y-3}{-2}=\frac{z-4}{5}$

(D) $\frac{x-2}{1}=\frac{y-3}{2}=\frac{z-4}{5}$

Q.5

The planes $3x-y+z+1=0, 5x+y+3z=0$ intersect in the line PQ. The equation of the plane through the point $(2,1,4)$ and the perpendicular to PQ is

(A) $x+y-2z=5$

(B) $x+y+2z=-5$

(C) $x+y+2z=5$

(D) $x+y-2z=-5$

Q.6

If from a point $P(a,b,c)$ perpendiculars PA and PB are drawn to yz and zx planes, then the equation of the plane OAB is

(A) $bcx+cay+abz=0$

(B) $bcx+cay-abz=0$

(C) $bcx-cay+abz=0$

(D) $-bcx+cay+abz=0$

Q.7

Under what condition do the planes $bx-ay=n, cy-bz=l, az-cx=m$ intersect in a line?

(A) $a+b+c=0$

(B) $a=b=c$

(C) $al+bm+cn=0$

(D) $l+m+n=0$

Q.8

A variable plane which remains at a constant distance $3\mathrm{p}$ from the origin cut the coordinate axes at A, B and C. The locus of the centroid of triangle ABC is

(A) $x^{-1}+y^{-1}+z^{-1}=p^{-1}$

(B) $x^{-2}+y^{-2}+z^{-2}=p^{-2}$

(C) $x+y+z=p$

(D) $x^2+y^2+z^2=p^2$

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Q.9

The equation of a plane passing through the line of intersection of the planes $x + 2y + 3z = 2$ and $x - y + z = 3$ and
at a distance $\frac{2}{\sqrt{3}}$ from the point $(3, 1, -1)$ is

(A) $5x - 11y + z = 17$

(B) $\sqrt{2}x + y = 3\sqrt{2} - 1$

(C) $x + y + z = \sqrt{3}$

(D) $x - \sqrt{2}y = 1 - \sqrt{2}$

Q.10

A plane passes through a fixed point (a, b, c) The locus of the foot of the perpendicular to it from the origin is the sphere

(A) $x^2 + y^2 + z^2 - ax - by - cz = 0$

(B) $x^2 + y^2 + z^2 - 2ax - 2by - 2cz = 0$

(C) $x^2 + y^2 + z^2 - 4ax - 4by - 4cz = 0$

(D) None of these

Q.11

The d.r. of normal to the plane through $(1, 0, 0), (0, 1, 0)$ which makes an angle $\pi/4$ with plane $x + y = 3$ are

(A) $(1, \sqrt{2}, 1)$

(B) $(1, 1, \sqrt{2})$

(C) $(1, 1, 2)$

(D) $(\sqrt{2}, 1, 1)$

Q.12

What is the distance between the planes $x - 2y + z - 1 = 0$ and $-3x + 6y - 3z + 2 = 0$?

(A) 3 unit

(B) 1 unit

(C) 0

(D) None of the above

Q.13

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What is the equation of the plane through z -axis and parallel to the line $\frac{x-1}{\cos\theta}=\frac{y+2}{\sin\theta}=\frac{z-3}{0}$?

(A) $x\cot\theta+y=0$

(B) $x\tan\theta-y=0$

(C) $x+y\cot\theta=0$

(D) $x-y\tan\theta=0$

Q.14

Two system of rectangular axes have the same orign. If a plane cuts them at distances $\mathrm{a}, \mathrm{b}, \mathrm{c}$ and $\mathrm{a}', \mathrm{b}', \mathrm{c}'$ respectively from the origin, then $\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=k\left(\frac{1}{a'^2}+\frac{1}{b'^2}+\frac{1}{c'^2}\right)$, where k is equal to

(A) 1

(B) 2

(C) 4

(D) None of these

Q.15

The locus of a point, such that the sum of the squares of its distances from the planes $x+y+z=0, x-z=0$ and $x-2y+z=0$ is 9, is

(A) $x^2+y^2+z^2=3$

(B) $x^2+y^2+z^2=6$

(C) $x^2+y^2+z^2=9$

(D) $x^2+y^2+z^2=12$

Q.16

The equation of the plane which makes with co-ordinate axes, a triangle with its centroid (α, β, γ) is

(A) $\alpha x+\beta y+\gamma z=3$

(B) $\alpha x+\beta y+\gamma z=1$

(C) $\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=3$

(D) $\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=1$

Q.17

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The direction ratios of the normal to the plane passing through the points $(1, -2, 3), (-1, 2, -1)$ and parallel to $\frac{x-2}{2} = \frac{y+1}{3} = \frac{z}{4}$ is

(A) $(2, 3, 4)$

(B) $(14, 0, 7)$

(C) $(-2, 0, -1)$

(D) $(2, 0, -1)$

Q.18

The plane $x + 3y + 13 = 0$ passes through the line of intersection of the planes $2x - 8y + 4z = p$ and $3x - 5y + 4z + 10 = 0$. If the plane is perpendicular to the plane $3x - y - 2z - 4 = 0$, then the value of p is equal to

(A) 2

(B) 5

(C) 9

(D) 3

Q.19

What is the angle between the planes $2x - y + z = 6$ and $x + y + 2z = 3$?

(A) $\pi/2$

(B) $\pi/3$

(C) $\pi/4$

(D) $\pi/6$

Q.20

What are the direction ratios of the line determined by the planes $x - y + 2z = 1$ and $x + y - z = 3$?

(A) $(-1, 3, 2)$

(B) $(-1, -3, 2)$

(C) $(2, 1, 3)$

(D) $(2, 3, 2)$

Answers & Solutions

Q.1 Answer:

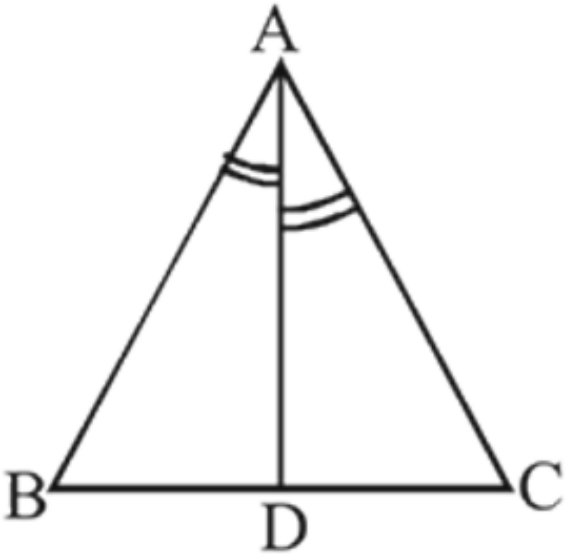
$\frac{2}{3}\sqrt{34}$

Solution:

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$AB = 6, BC = \sqrt{13}, CA = 3$

$\therefore AB : AC = 2 : 1$



Internal bisector of an angle divides the opposite side in the ratio of adjacent sides

$\therefore \quad \frac{BD}{CD} = \frac{AB}{AC} = \frac{2}{1}$

$\therefore$ **Coordinate of** D **are** $\left(2, \frac{13}{3}, 6\right)$

$\therefore \quad$ **Length** $AD = \frac{2}{3}\sqrt{34}$

Q.2 Answer:

$x^2 + y^2 + z^2 = 9$

Solution:

Let the variable point be (α, β, γ) **then according to question** $\left(\frac{|\alpha+\beta+\gamma|}{\sqrt{3}}\right)^2 + \left(\frac{|\alpha-\gamma|}{\sqrt{2}}\right)^2 + \left(\frac{|\alpha-2\beta+\gamma|}{\sqrt{6}}\right)^2 = 9$

$\Rightarrow \alpha^2 + \beta^2 + \gamma^2 = 9.$

So, the locus of the point is $x^2 + y^2 + z^2 = 9$

Q.3 Answer:

$x^2 + y^2 + z^2 - x - 2y - 3z = 0$

Solution:

Let $P(\alpha, \beta, \gamma)$ **be the foot of the perpendicular from the origin** $O(0, 0, 0)$ **to the plane So, the plane passes through** $P(\alpha, \beta, \gamma)$ **and is perpendicular to OP. Clearly direction ratios of OP i.e., normal to the plane are** α, β, γ**. Therefore, equation of the plane is** $\alpha(x - \alpha) + \beta(y - \beta) + \gamma(z - \gamma) = 0$

This plane passes through the fixed point $(1, 2, 3)$**, so** $\alpha(1 - \alpha) + \beta(2 - \beta) + \gamma(3 - \gamma) = 0$ **or** $\alpha^2 + \beta^2 + \gamma^2 - \alpha - 2\beta - 3\gamma = 0$

Generalizing α, β **and** γ**, locus of** $P(\alpha, \beta, \gamma)$ **is** $x^2 + y^2 + z^2 - x - 2y - 3z = 0$

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Q.4 Answer:

$\frac{x-2}{1} = \frac{y-3}{-2} = \frac{z-4}{5}$

Solution:

Let Q be the image of the point $\mathrm{P}(2,3,4)$ in the plane $\mathrm{x}-2\mathrm{y}+5\mathrm{z}=6$, then PQ is normal to the plane $\therefore$ direction ratios of PQ are $<1,-2,5>$ Since PQ passes through $\mathrm{P}(2,3,4)$ and has direction ratios $1,-2,5$

$\therefore$ Equation of PQ is $\frac{\mathrm{x}-2}{1} = \frac{\mathrm{y}-3}{-2} = \frac{\mathrm{z}-4}{5}$

Q.5 Answer:

$x+y-2z=-5$

Solution:

Let $\{l,\ \mathrm{m}, \mathrm{n}\}$ be the direction -cosines of PQ, then $3l-m+n=0$ and $5l+m+3n=0$

$\therefore \frac{l}{-3-1} = \frac{\mathrm{m}}{5-9} = \frac{\mathrm{n}}{3+5}$ i.e $\frac{l}{1} = \frac{\mathrm{m}}{1} = \frac{\mathrm{n}}{-2}$

Now a plane $\perp$ to PQ will have l, m, n as the coefficients of x, y and z

Hence the plane $\perp$ to PQ is $\mathrm{x}+\mathrm{y}-2\mathrm{z}=\lambda$ It passes through $(2,1,4); \therefore 2+1-2.4=\lambda$ i.e $\lambda=-5$ Hence the required plane is $x+y-2z=-5$

Q.6 Answer:

$bcx+cay-abz=0$

Solution:

$A(0,b,c)$ in yz -plane and $B(a,0,c)$ in zx -plane. Plane through O is $px+qy+rz=0$. It passes through A and $B. \therefore 0p+qb+rc=0$ and $pa+0q+rc=0$

$\Rightarrow \frac{p}{bc} = \frac{q}{ca} = \frac{r}{-ab} = k$

$\Rightarrow p=bck, q=cak$ and $r=-abk.$

Hence required plane is $bcx+cay-abz=0.$

Q.7 Answer:

$al+bm+cn=0$

Solution:

The planes $bx-ay=n, cy-bz=l$ and $az-cx=m$ intersect in a line, if $al+bm+cn=0.$

Q.8 Answer:

$x^{-2}+y^{-2}+z^{-2}=p^{-2}$

Solution:

Let equation of the variable plane be $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ This meets the coordinate axes at $\mathrm{A}(\mathrm{a},0,0), \mathrm{B}(0,\ \mathrm{b}, 0)$ and $\mathrm{C}(0,0,\mathrm{c})$.

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Let $\mathrm{P}(\alpha, \beta, \gamma)$ be the centroid of the $\triangle \mathrm{ABC}$. Then $\alpha = \frac{\mathrm{a}+0+0}{3}, \beta = \frac{0+\mathrm{b}+0}{3}, \gamma = \frac{0+0+\mathrm{c}}{3}$

$\therefore \mathrm{a} = 3\alpha, \mathrm{b} = 3\beta, \mathrm{c} = 3\gamma$

Plane (1) is at constant distance $3\mathrm{p}$ from the origin, so $3p = \frac{\left|\frac{0}{a}+\frac{0}{b}+\frac{0}{c}-1\right|}{\sqrt{\left(\frac{1}{a}\right)^2+\left(\frac{1}{b}\right)^2+\left(\frac{1}{c}\right)^2}} \Rightarrow \frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2} = \frac{1}{9p^2}$

From (2) and (3), we get

$\frac{1}{9\alpha^2}+\frac{1}{9\beta^2}+\frac{1}{9\gamma^2} = \frac{1}{9\mathrm{p}^2}$

$\Rightarrow \alpha^{-2}+\beta^{-2}+\gamma^{-2} = \mathrm{p}^{-2}$

Generalizing α, β, γ, locus of centroid $\mathrm{P}(\alpha, \beta, \gamma)$ is $\mathrm{x}^{-2}+\mathrm{y}^{-2}+\mathrm{z}^{-2} = \mathrm{p}^{-2}$

Q.9 Answer:

$5x - 11y + z = 17$

Solution:

The plane passing through the intersection line of given planes is

$$(x + 2y + 3z - 2) + \lambda(x - y + z - 3) = 0$$

or $(1+\lambda)x + (2-\lambda)y + (3+\lambda)z + (-2-3\lambda) = 0$

Its distance from the point $(3, 1, -1)$ is $\frac{2}{\sqrt{3}}$

$$\therefore \left|\frac{3(1+\lambda)+1(2-\lambda)-1(3+\lambda)+(-2-3\lambda)}{\sqrt{(1+\lambda)^2+(2-\lambda)^2+(3+\lambda)^2}}\right| = \frac{2}{\sqrt{3}}$$

$$\Rightarrow \left|\frac{-2\lambda}{\sqrt{3\lambda^2+4\lambda+14}}\right| = \frac{2}{\sqrt{3}}$$

$$\Rightarrow 3\lambda^2 + 4\lambda + 14 = 3\lambda^2 \Rightarrow \lambda = -\frac{7}{2}$$

$\therefore$ Required equation of plane is

$$(x + 2y + 3z - 2) - \frac{7}{2}(x - y + z - 3) = 0$$

or $5x - 11y + z = 17$

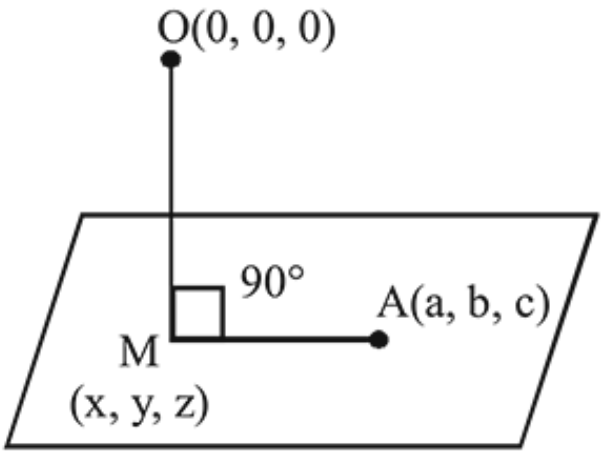
Q.10 Answer:

$x^2 + y^2 + z^2 - ax - by - cz = 0$

Solution:

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Let $A(a, b, c)$ be the fixed point on the variable plane



NowD. R 's of OM are $x-0, y-0, z-0$ i.ex, y, z
D.R.'s of MA are $x-a, y-b, z-c$.
Since OM perpendicular MA
$x(x-a)+y(y-b)+z(z-c)=0$
$\Rightarrow x^2+y^2+z^2-ax-by-cx=0$

Q.11 Answer:

$(1,1,\sqrt{2})$

Solution:

Equation of plane through $(1,0,0)$ is $a(x-1)+by+cz=0$
(i) passes through $(0,1,0)$. $-a+b=0 \Rightarrow b=a$; Also,
$\cos 45^\circ=\frac{a+a}{\sqrt{2(2a^2+c^2)}} \Rightarrow 2a=\sqrt{2a^2+c^2}$
$\Rightarrow 2a^2=c^2 \Rightarrow c=\sqrt{2}a.$
So d.r of normal are a, a $\sqrt{2}a$ i.e. $1, 1, \sqrt{2}$.

Q.12 Answer:

None of the above

Solution:

Given planes are
$x-2y+z=1 \quad \ldots.$
and $-3x+6y-3z=-2$
$\equiv x-2y+z=\frac{2}{3}$
Since both planes are parallel and $a=1,\ b=-2, c=1$
and $d_1=-1,\ d_2=\frac{-2}{3}$

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$\therefore$ **Distance** $= \left|\frac{d_2-d_1}{\sqrt{a^2+b^2+c^2}}\right|$

Distance $= \left|\frac{1-\frac{2}{3}}{\sqrt{1+4+1}}\right| = \frac{1}{3\sqrt{6}}$

Q.13 Answer:

$x\tan\theta - y = 0$

Solution:

Let equation of plane through z-axis is

$ax + by = 0$

It is given that this plane is parallel to the line

$\frac{x-1}{\cos\theta} = \frac{y+2}{\sin\theta} = \frac{z-3}{0}$

Since the plane parallel to the line

$\therefore \quad a\cos\theta + b\sin\theta = 0$

$\Rightarrow a\cos\theta = -b\sin\theta \Rightarrow a = -b\tan\theta$

$\therefore \quad -b\tan\theta x + by = 0$

$\Rightarrow \quad x\tan\theta - y = 0 (\because b \neq 0)$

which is required equation of plane.

Q.14 Answer:

1

Solution:

Let a, b, c **be the intercepts when** Ox, Oy, Oz **are taken as axes, then the equation of the plane is**

$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

let a, b, c **be the intercepts when** (OX, OY, OZ) **are taken as axes; then in this case equation of the same plane is** $\frac{X}{a} + \frac{X}{b} + \frac{X}{c} = 1$

Now, eqs. (i) and (ii) are equations of the same plane and in both the cases the origin is same. Hence, length of the perpendicular drawn from the origin to the plane in both the case must be the same

$\frac{1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}} = \frac{1}{\sqrt{\frac{1}{a^2}+\frac{1}{b'^2}+\frac{1}{c'^2}}}$

$\Rightarrow \quad \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{a'^2} + \frac{1}{b'^2} + \frac{1}{c'^2}$

$k = 1$

Q.15 Answer:

$x^2 + y^2 + z^2 = 9$

Solution:

Let the variable point be (α, β, γ) **then according to question**

$\left(\frac{|\alpha+\beta+\gamma|}{\sqrt{3}}\right)^2 + \left(\frac{|\alpha-\gamma|}{\sqrt{2}}\right)^2 + \left(\frac{|\alpha-2\beta+\gamma|}{\sqrt{6}}\right)^2 = 9$

$\Rightarrow \alpha^2 + \beta^2 + \gamma^2 = 9$

So, the locus of the point is $x^2 + y^2 + z^2 = 9$

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Q.16 Answer:

$\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=3$

Solution:

(c) Let us take a triangle ABC **and their vertices** $\mathrm{A}(\mathrm{a}, 0,0), \mathrm{B}(0,\ \mathrm{b}, 0)$ **and** $\mathrm{C}(0,0, \mathrm{c})$

Therefore the equation of plane is

$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$

Now, given centroid of $\Delta \mathrm{ABC}$ **is** (α, β, γ) **As we know, centroid of** $\triangle \mathrm{ABC}$ **with vertices** $\left(x_1, y_1, z_1\right),\left(x_2, y_2, z_2\right)$ **and** $\left(x_3, y_3, z_2\right)$ **is given by**

$$\left(\frac{\mathrm{x}_1+\mathrm{x}_2+\mathrm{x}_3}{3}, \frac{\mathrm{y}_1+\mathrm{y}_2+\mathrm{y}_3}{3}, \frac{\mathrm{z}_1+\mathrm{z}_2+\mathrm{z}_3}{3},\right)$$

$\therefore$ **By using this formula, we have** $\frac{a+0+0}{3}=\alpha \Rightarrow a=3 \alpha,; \frac{0+b+0}{3}=\beta$

$\Rightarrow \mathrm{b}=3 \beta$

and $\frac{0+0+c}{3}=\gamma \Rightarrow c=3 \gamma$

Now, put the values of $\mathrm{a}, \mathrm{b}, \mathrm{c}$ **in equation (i), which gives**

$\frac{x}{3 \alpha}+\frac{y}{3 \beta}+\frac{z}{3 \gamma}=1 \quad \therefore \frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=3$

Q.17 Answer:

$(2,0,-1)$

Solution:

Any plane through $(1,-2,3)$ **is** $\mathrm{A}(\mathrm{x}-1)+\mathrm{B}(\mathrm{y}+2)+\mathrm{C}(\mathrm{z}-3)=0$

The point $(-1,2,-1)$ **lies in this plane if** $-2 \mathrm{~A}+4 \mathrm{~B}-4 \mathrm{C}=0$

i.e. if $\mathrm{A}-2 \mathrm{~B}+2 \mathrm{C}=0$

The plane (1) is parallel to the given line with d.r. $2,3,4$

if $2 \mathrm{~A}+3 \mathrm{~B}+4 \mathrm{C}=0$

From (2) and (3), we have $\frac{\mathrm{A}}{-8-6}=\frac{\mathrm{B}}{4-4}=\frac{\mathrm{C}}{3+4}$

$\Rightarrow \frac{\mathrm{A}}{-14}=\frac{\mathrm{B}}{0}=\frac{\mathrm{C}}{7} \Rightarrow \mathrm{A}: \mathrm{B}: \mathrm{C}=2: 0:-1$

Q.18 Answer:

3

Solution:

Equation of plane passes through line of intersection of the plane $2x-8y+4z=p$ **and** $3x-5y+4z+10=0$ **is**

$(2x-8y+4z-p)+\lambda(3x-5y+4z+10)=0$

$\Rightarrow(2+3\lambda)x+(-8-5\lambda)y+(4+4\lambda)z-p+10\lambda=0$

Given equation of plane is

$x+3y+13=0 \quad \ldots.(\mathrm{ii})$

Equations (i) and (ii) represent same plane.

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$\therefore 4+4\lambda=0 \Rightarrow \lambda=-1$

Putting $\lambda=-1$ in equation (i), we get

$$-\mathrm{x}-3\mathrm{y}-\mathrm{p}-10=0$$

$$\Rightarrow \mathrm{x}+3\mathrm{y}+\mathrm{p}+10=0 \quad \ldots(\mathrm{iii})$$

Comparing the coefficient of x, y and z constants term for equations (i) and (ii), we get

$$\mathrm{p}+10=13$$

$$\Rightarrow \mathrm{p}=3$$

Q.19 Answer:

$\pi/3$

Solution:

We know, if $a_1x+b_1y+c_1z=d_1$ and $a_2x+b_2y+c_2z=d_2$ are two planes then angle between them is $\cos\theta=\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}$

Let q be the angle between given planes Here, $a_1=2, b_1=-1, c_1=1; a_2=1, b_2=1, c_2=2$

$\cos q=\frac{2\times1+1\times(-1)+1\times2}{\sqrt{4+1+1}\sqrt{1+1+4}}=\frac{3}{6}=\frac{1}{2}=\cos\frac{\pi}{3}$

$\Rightarrow \theta=\frac{\pi}{3}$

Q.20 Answer:

$(-1,3,2)$

Solution:

The intersection of given plane is $x-y+2z-1+\lambda(x+y-z-3)=0$

$\Rightarrow x(1+\lambda)+y(\lambda-1)+z(2-\lambda)-3\lambda-1=0$

DR's of normal to the above plane is $(1+\lambda, \lambda-1, 2-\lambda)$

By taking option (a)

$-1(1+\lambda)+3(\lambda-1)+2(2-\lambda)=0$

$\Rightarrow -1-\lambda+3\lambda-3+4-2\lambda=0$

$\Rightarrow 0=0$ **which is true.**

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